WS 2026/27 · Hochschule Düsseldorf

AI & Computer Vision meets Creation

Semester plan: topics, format, dates

Fridays · 10:00–13:00 · Lectures in German, materials in English

How the semester works

01

Micro Deep Dives

In pairs, you research one frontier topic and present it to the class.

02

Design Thinking

One full session before Christmas. Form your hackathon teams and frame your challenge: what must your system see?

03

Hackathon

One full day. Build, break, learn, explain. Graded at the end.

Topics

01
CV foundations: images as data

02
Deep learning for vision: CNNs & Vision Transformers

03
Object detection & evaluation

04
Multi-object tracking

05
Pose estimation & gesture

06
Depth & 3D perception

07
Foundation models: SAM & open-vocabulary vision

08
Generative AI & synthetic data

09
Bias, ethics & the EU AI Act

Plus Micro Deep Dives on frontier topics, Gaussian Splatting, Open-Vocabulary Vision, World Models, Vision-Language Models, Monocular Depth, Edge AI

Schedule · October–November

Friday	Session
16.10.	Kick-off · CV foundations
23.10.	Deep learning for vision · gesture recognition
30.10.	Bias & deepfakes
06.11.	Object detection · Deep Dive kick-off: topics & pairs
13.11.	Deep Dive studio · coaching on site
20.11.	Deep Dive work in pairs · no on-site session
27.11.	Deep Dive work in pairs · no on-site session

All sessions Fridays 10:00–13:00 unless noted

Schedule · December–January

Friday	Session
04.12.	Deep Dive presentations I · two pairs · Tracking
11.12.	Deep Dive presentations II · one pair · Depth & 3D
18.12.	Design Thinking Workshop · hackathon teams & challenges
23.12.–03.01.	Christmas break · no lectures
08.01.	Remote session · hackathon plan review
15.01.	Foundation models & diffusion · hackathon check-in
22.01.	Hackathon · full day, graded

All sessions Fridays 10:00–13:00 unless noted

Mark your calendar

20.11. + 27.11.

No on-site session. Deep Dive work in pairs.

04.12. + 11.12.

Deep Dive presentations. Slides due 03.12. for all pairs.

08.01.

Remote session. Link follows.

22.01.

Hackathon. Full day, on site. Location & times: [tbd], 09:00-17:30

Micro Deep Dives

What it is

In pairs, you research one frontier topic in Computer Vision and teach it to the class. You go deeper than the lectures, on something the field is moving on right now.

Format

- 35–40 minutes: about 15–20 min theory, 15–20 min hands-on activity for the class
- German or English, your choice
- Any format: slides, whiteboard, live demo
- Full reference list (APA or IEEE)
- Both partners present and answer questions

Deep Dive topics

Gaussian Splatting

How do a few photos become a 3D scene? How is it different from NeRF?

Open-Vocabulary Vision

How do models find objects without a fixed class list? What can SAM do that YOLO can't?

World Models

How does a model learn what happens next? Why do robots and self-driving cars need one?

Vision-Language Models

How do models connect seeing and language? Where do they hallucinate?

Monocular Depth

How can a single photo give depth without LiDAR? Where does it fail?

Edge AI

How do vision models run on a phone or Raspberry Pi? What do you lose in accuracy?

One topic per pair. Deepfakes were covered on 30.10., diffusion models follow on 15.01.

Deep Dive timeline

Date	What
06.11.	Kick-off: choose topic and partner
13.11.	Studio session: coaching per pair, on site
20.11. + 27.11.	Independent work, no on-site session
03.12.	Slides due, all pairs
04.12.	Presentations: two pairs
11.12.	Presentation: third pair, ideally a 3D-related topic

The Starter Kit has a tutor prompt you can paste into any LLM to plan your preparation.

Grading & AI use

Grading

Up to 0.3–0.5 on your final grade

Not formally graded. What counts: technical depth, clear structure, a meaningful activity for the class, your own examples and reflection, and teamwork.

Using AI

AI supports you, it doesn't think for you

Use it for research, structure, code or images, but understand everything you present. No unchecked text or visuals. Note what you used AI for, e.g. "ChatGPT, structuring".

Evaluation criteria

Dimension	What we look for
Problem clarity	Is the problem concrete, relevant, and well framed?
CV relevance	Is vision essential, not replaceable by rules or text?
Technical realism	Is the scope plausible and well reasoned?
Reflection & limits	Are assumptions, risks, and failure modes explicit?
Narrative structure	Is there a clear and logical story?
Visual reasoning	Do visuals support understanding and thinking?
Clarity & confidence	Is the presentation clear, calm, and convincing?
Shared ownership	Are all team members meaningfully involved?
Quality of insight	Does the project offer a thoughtful or novel perspective?
Engagement with complexity	Does the team engage honestly with uncertainty?

Plus: Micro Deep Dive and personal documentation, due 29.01.2027